COVID Difficult Intubation Strategies Case

Section 1: Case Summary

Scenario Title:	**COVID Difficult Intubation Strategies**
Keywords:	COVID-19, laryngoscopy, difficult airway
Brief Description of Case:	A 60-year-old male with obvious head injury and respiratory failure arrives to the ED in C-spine precautions during the COVID-19 pandemic. Because of the c-spine immobilization and COVID precautions, the team tries hyperangulated video laryngoscopy in their initial airway attempt. However, this attempt will not be successful and the patient will desaturate, necessitating bag-valve masking. At this point, ventilation will become difficult, even with optimized two-person BVM technique or placement of a supraglottic airway. Team leader will be required to exhaust the entire difficult airway algorithm, finally securing an airway with a surgical approach.

Goals and Objectives	
Educational Goal:	Work through intubation strategies for a patient in C-spine precautions while taking COVID-19 precautions.
Objectives: (Medical and CRM)	1. Demonstrate proper donning and doffing for a COVID-precaution intubation 2. Recognize the need for early intubation in a trauma patient 3. Communicate a stepwise airway plan for a potentially difficult airway 4. Demonstrate a successful transition through a difficult airway algorithm 5. Recognize and proceed through a stepwise plan when confronted with a “can’t intubate can’t oxygenate” scenario
EPAs Assessed:	C3: Providing airway management and ventilation for a predicted difficult airway.

Learners, Setting and Personnel				
Target Learners:	☒ Junior Learners	☒ Senior Learners	☒ Staff	
	☒ Physicians	☒ Nurses	☒ RTs	☒ Inter-professional
	☐ Other Learners:			
Location:	☒ Sim Lab	☒ In Situ	☐ Other:	
Recommended Number of Facilitators:	Instructors: 1			
	Sim Actors: 1-2			
	Sim Techs: 1			

Scenario Development	
Date of Development:	July 7, 2020
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Revised By:	
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Mr. Johnson			Age: 60	Gender: Male	Weight: 70 kg
Presenting complaint: Head injury and Respiratory failure					
Temp: 38.0	HR: 80	BP: 100/65	RR: 20	O_2Sat: 80% (RA) 95% (NRB)	FiO_2: 100%
Cap glucose: 7			GCS: 8 (E1 V2 M3)		
Triage note: 60-year-old male found at the bottom of 3 steps at home by wife. GCS 6 with obvious bruising to head. C-spine collar on. Patient was swabbed for COVID yesterday due to a new cough and fever and the results are still pending.					
Allergies: none					
Past Medical History: COPD			Current Medications: spiriva, ventolin		

Section 2B: Extra Patient Information

A. Further History
Wife/EMS: patient had been coughing over the past several days with increased sputum production and purulence. She hadn't heard the patient for 20 minutes, so went looking for him and found him altered at the bottom of three steps inside their house.

B. Physical Exam	
List any pertinent positive and negative findings	
Cardio: normal S1+S2, no murmurs, no leg edema	Neuro: GCS 6, PEARL 3 mm
Resp: wheezes throughout, crackles to right lower lobe	Head & Neck: large bruise to forehead, c-spine collar on
Abdo: soft, non-tender, bowel sounds present	MSK/skin: normal
Other: No other signs of trauma	

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin *(adult)*
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid

B. Special Equipment Required
PPE- gowns, gloves, face shields, N95 masks (expired or simulated) IV supplies Nasal cannula Face masks Bag-vale mask with viral filter and PEEP valve Video laryngoscopy (glidescope machine, introducer, stylet) Direct laryngoscopy (size 3 macintosh, stylet) End tidal CO2 Ventilator **Difficult airway cart/equipment** Bougie Supraglottic airway device (eg. LMA) 11 blade scalpel, 6.0 endotracheal tube

C. Required Medications
RSI kit Vasopressors

D. Moulage
C-spine collar, Bruising to head

E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor

F. Patient Reactions and Exam
Decreased air entry and wheezes throughout with crackles to right lower lobe. Only moaning. Does not open eyes.

Section 4: Sim Actors and Standardized Patients

Sim Actors and Standardized Patient Roles and Scripts	
Role	*Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)*
Bedside RNs	Skillful with critical care tasks. Helpful team member.
Resp Therapist	Skillful with critical care tasks. Helpful team member.

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: sinus HR: 80 BP: 100/65 RR: 20 O_2SAT: 95%(NRB) T: 38.0°C GCS: 6	*GCS 6 with signs of head injury and c-spine collar* *Respiratory distress and fever*	Expected Learner Actions ☐ Dons appropriate PPE ☐ 2 peripheral IVs, Monitors ☐ Oxygenation with nasal cannula + non-rebreather mask ☐ Trauma blood work ☐ Primary trauma survey ☐ Order CXR, pelvis x-ray ☐ Order CT head and c-spine	Modifiers - None Triggers Completion of all actions or after 5 minutes → Next State	Learner must appropriately don PPE prior to entering the room.
2. Declining Status Rhythm: sinus HR: 80 BP: 100/65 RR: 30 O_2SAT: 88%(NRB)	*GCS 6 with signs of head injury and c-spine collar*	Expected Learner Actions ☐ Recognize need for intubation ☐ Calls for 2nd physician/RN/RT for backup outside the room ☐ Dons airborne PPE ☐ Discusses intubation plan with team, including Plan A+B+C ☐ Orders pressors, induction and paralytic agents ☐ Verbalizes O2 sat threshold for starting BVM for patient	Modifiers - None Triggers Paralytic given → Next State	If they have not already, the learner must don an N-95 respirator.
3. Attempt Intubation Rhythm: sinus HR: 80 BP: 145/95 RR: 30 O_2SAT: 88% (NRB)		☐ Manual c spine immobilization ☐ Wait 60 seconds for paralysis ☐ Attempt intubation ☐ Optimize view ☐ Cricoid pressure	Modifiers - Post Paralytic: O_2 sats→80% - During intubation-: O_2 sats→70% Triggers Learner attempts laryngoscopy, but cannot achieve adequate view → Next State	Some mannequins can be set up so that it is not possible to intubate with video laryngoscopy **or** the team leader can be told verbally that intubation with video laryngoscope is not possible

4. Attempt Rescue Plan Rhythm: sinus HR: 110 BP: 100/65 RR: 0 (paralyzed) O_2SAT: 70%	*Paralyzed*	☐ Continue manual c-spine immobilization ☐ Insert oral airway ☐ 2-person BVM + PEEP valve ☐ Place supraglottic airway ☐ Attempt Plan B intubation (eg. direct laryngoscopy + bougie) ☐ Asks 2nd physician to enter the room	<u>Modifiers</u> Prolonged attempts: O_2 sat→60% <u>Triggers</u> Calls for Cric Kit → Next State	Pt is difficult to bag despite 2-person BVM technique or SGA Unable to visualize cords with direct laryngoscopy Unable to feel tracheal rings with blind bougie placement
5. CICO/CICV Rhythm: sinus HR:110 BP: 100/65 RR: 0 (bagged) O2 SAT: 60%	*Paralyzed*	☐ Ask outside team member to call anesthesia/ENT ☐ Perform Cricothyrotomy (scalpel-bougie method) ☐ Doffs PPE safely	<u>Modifiers</u> Post-Cric: O_2 sat→90% <u>Triggers</u> Cric placed → PPE Doffed→ **END CASE**	ETT is placed via cric, easy to bag, sats improve

Appendix A: Laboratory Results

None given in case	

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

None given in case

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

1. The purpose of this case is to have learners prepare for multiple intubation strategies in a patient with an anatomically difficult airway while taking COVID-19 precautions.

2. The learner must be able to properly don and doff for the patient assessment and for the airway management, specifically.

3. Although the patient is having a COPD exacerbation, management of this is not the primary learning objective of this case. Preparing and executing the Plan A & B & C of the intubation plan are the main learning issues.

4. Given the anatomically difficult airway (c-spine immobilization) and the COVID intubation strategy of prioritizing video intubation methods, the learner should first attempt video laryngoscopy (in our center- we use hyperangulated VL systems). When this attempt fails, they must trouble shoot and use alternative strategies.

5. Scenario is designed to push the team leader through their entire airway plan.
Unable to intubation with laryngoscopy → multiple approaches to ventilation of the patient unsuccessful (OPA, 2-person BVM, supraglottic device) → proceed to "can't intubate can't oxygenate" pathway necessitating a cricothyrotomy

<u>Discussion points</u>
This case is specifically designed to practice a scenario where an RSI performed under covid conditions is complicated by inability to intubate the patient and having the patient desaturate to a dangerous level.

Points to discuss can include:
- when to initiate bagging the patient (as most algorithms advocate avoiding bagging patients if possible)
- how to coordinate with the team assisting outside the closed room
- how to get additional help (from eg. Anesthesia)
- preparing necessary extra equipment (LMA, surgical airways)

References

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4. https://cts-sct.ca/wp-content/uploads/2020/05/CJRCCSM_Addressing-therapeutic-questions-to-optimize-COPD-management-during-the-COVID-19-pandemic.pdf